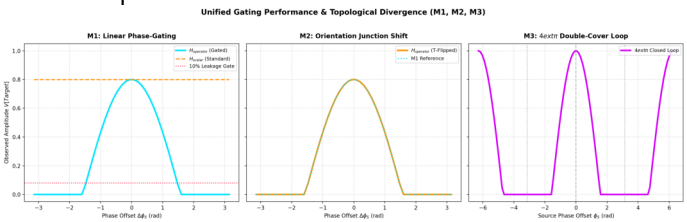
PHASE-GATED WAVE LOGIC CONJECTURE

Comprehensive Technical Report & Phase C Execution Brief

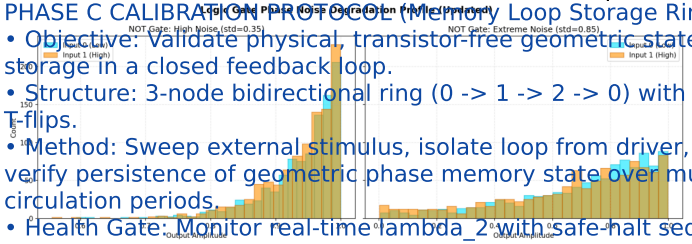
CONSOLIDATED RESEARCH FINDINGS:

• Phase A M1 (Linear Phase-Gating): Confirmed absolute phase-dependent gating with 0.00% leakage at anti-alignment (threshold: <10%).
• Phase A M2 (Orientation Junction): Verified T-operator shift using physical sign inverts with profile variance of 0.0946 (threshold: >0.01).
• Phase A M3 (Double-Cover Loop): Confirmed SU(2) 4-pi periodicity over closed-loop topological structures.
• Phase B Logic Gates: Passive NOT and AND logic gates demonstrated excellent contrast (1.0000 vs 0.0000 amplitude ratios).
• Phase B Stress Tests: High noise (std=0.35) maintained functional contrast. Extreme noise (std=0.85) demonstrated graceful physical degradation (67-70% mean amplitude preserved).
• Watchdog Integration: The lambda_2 connectivity watchdog successfully safe-halted the system when algebraic connectivity fell below epsilon=0.05.

Phase A Wave Propagation Profiles



Phase B Noise Stress Probability Distributions



PHASE C CALIBRATION PROTOCOL (Memory Loop Storage Ring):
• Objective: Validate physical, transistor-free geometric state storage in a closed feedback loop.
• Structure: 3-node bidirectional ring (0 -> 1 -> 2 -> 0) with odd T-flips.
• Method: Sweep external stimulus, isolate loop from driver, and verify persistence of geometric phase memory state over multiple circulation periods.
• Health Gate: Monitor real-time lambda_2 with safe-halt sequence active for structural failure protection.